

NAVYA BHAT

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TECHNICAL SKILLS

Languages: Python, SQL, Java, C++, JavaScript, C

Machine Learning: PyTorch, TorchVision, TensorFlow/Keras, scikit-learn, XGBoost, pandas, OpenCV, NumPy, Matplotlib, Ultralytics

Tools: Git, Docker, Google Cloud Platform, REST APIs, Streamlit, React, Node.js, Flask, Tableau, Alteryx

Systems: UNIX/Linux, kernel programming, concurrency, GDB, Valgrind, Bash, GNU Make

EDUCATION

University of Southern California

Los Angeles, CA

M.S. Computer Science, GPA 4.0/4.0

August 2024 – May 2027 (expected)

Key courses: Machine Learning, Algorithms, Operating Systems, Information Retrieval, Web Technologies (Full Stack)

Kellogg School of Management, Northwestern University

Evanston, IL

M.B.A., GPA 3.53/4.0; Kellogg Growth Scholarship (\$100K); Data Analytics and Finance focus

August 2018 – June 2020

Institute of Chemical Technology

Mumbai, India

B.E. in Chemical Engineering, GPA 8.45/10; Sir Ratan Tata Scholarship (top 5% of class)

July 2010 – May 2014

WORK EXPERIENCE

Lawrence Livermore National Laboratory · *Machine Learning Intern, NIF*

Livermore, CA · June 2026 – September 2026

- Built an end-to-end ML object detector for laser images from the National Ignition Facility's fusion experiments using an FCOS and YOLO11 ensemble with a ResNet crop verifier. Added DoG preprocessing to surface low-contrast defects, and designed location-aware cross-validation and ensemble voting to select the final system at 0.90 recall, cutting 3–4 hours of manual review per week
- Developed a reliable, end-to-end agentic workflow in Python using 6 AI agents that coordinated CT segmentation, 3D registration, strut-level defect scoring with uncertainty flagging, and reporting through explicit artifact handoffs; enabled agents to selectively rerun dependent stages and iterate on reviewer feedback through a chat-driven Streamlit dashboard

Starbucks · *Manager, Existing Store Strategy*

Seattle, WA · June 2023 – May 2024

- Shaped Starbucks' renovation strategy using a data-driven prioritization model, collaborating across product, marketing, and operations teams to create scalable reinvestment guidelines and provide strategic recommendations to leadership
- Shipped a SQL-backed Tableau dashboard that automated renovation tracking across multi-source data, saving 5+ hours weekly

Boston Consulting Group · *Consultant*

Chicago, IL · May 2021 – June 2023

- Designed and analyzed a large-scale consumer survey to build a quantitative growth strategy for a fast-food chain, matching product innovation to consumer demand patterns; informed a new store concept projected at \$15–20B annual revenue
- Delivered strategic pricing recommendations for an alcoholic beverage brand through competitive benchmarking, P&L analysis, and price-profitability modeling using demand elasticity forecasts; estimated an 11-point increase in gross margin
- Consolidated 8+ legacy pricing tools into 2 automated Alteryx decision-support workflows for a grocery wholesaler, improving system performance and streamlining pricing decisions

Dow Chemical Company · *Senior Account Executive*

Mumbai, India · June 2014 – May 2018

- Grew B2B revenue for packaging vertical from \$23M to \$45M in two years by optimizing price-volume strategy, strengthening client relationships and crafting technical value propositions; won Dow's Pinnacle Award for global selling excellence
- Led India's first recyclable edible-oil packaging launch to advance sustainable packaging, running feasibility analysis, ROI modeling, and supplier negotiations to add \$4M in annual revenue

PROJECTS

- Industrial fault detection.** Built a two-stage fault-detection pipeline on 52-variable Tennessee Eastman time-series data: LSTM and CNN binary detectors with validation-locked thresholds and consecutive- k alarm rules cut false alarms to near zero (CNN: 14% to 0% at 95% detection), and GNN, LSTM, CNN, and Random Forest classifiers diagnosed 20 fault types with up to 99% accuracy
- FathomNet 2026 Kaggle competition.** Built a PyTorch object detection pipeline for 32 marine species under domain shift: ensembled YOLOv8x and RT-DETR with multi-scale TTA and Weighted Boxes Fusion, relabeled detections via DINOv2 consensus, and selected MBARI's marine-imagery weights after benchmarking 5 pretrained initializations (placed 16th of 102 teams)
- Waste image classification.** Trained multi-class classifiers in TensorFlow/Keras on RealWaste (4.7K images, 9 classes) using pretrained CNNs (VGG16, ResNet50/101, EfficientNetB0); best model (ResNet50) reached 79% test accuracy and 0.98 AUC
- Predictive maintenance.** Built Random Forest and XGBoost ensembles on the APS truck failure dataset; handled severe class imbalance with SMOTE, applied feature transformations, and tuned hyperparameters for robustness
- Full-stack event platforms.** Built and deployed two event-discovery apps on GCP using React/Node.js/MongoDB and Python/Flask, integrating Ticketmaster, Spotify, and Google Maps APIs with autosuggest, geolocation, and favorites
- Multithreaded web crawler.** Crawled 10K news pages in Java with crawler4j using domain filtering, MIME normalization, and politeness delays; reported status distribution and efficiency